Rajarata University of Sri Lanka

Department of Computing

COM 3401 – Data Communication and Networking
Laboratory 05 – DHCP, DNS and Application Services
Time: 2 Hours

Intended Learning Outcomes (ILOs)

By the end of this practical, students should be able to:

1. Configure network devices to obtain addresses dynamically
2. Configure DNS

Pre-Practical Preparation

1. **Reading/Resources**: Refer to the topics covered in Lecture 3 and Lecture 4

Practical Objective:

Understand how client-server networking works. Students should be able to configure DHCP and thereby enable devices to obtain addresses dynamically. Students will learn to configure DNS and learn how DNS helps to resolve hostnames. Getting a basic introduction on providing FTP service using a server.

Practical Activity

Activity 1:

Designing the topology

- Objectives: Designing a topology which will enable the learning process of this whole practical session
- Instructions:
    1. Create a new project
    2. Add following devices:
        - 1 x switch (2960)
        - 3 x PC
        - 1 x Router (PT8200)
        - 1 x Server

3. Connect the PCs and the server to the switch
4. Connect switch (interface 0/0/0) to the router

Activity 2:

Configuring DHCP

- Objective: Making students aware on how to manually configure DHCP on a router
- Instructions:
    1. Configure DHCP as follows using router CLI
        - enable
        - configure terminal
        - ip dhcp excluded-address 192.168.10.1 192.168.10.10
        - ip dhcp pool CAMPUS
        - network 192.168.10.0 255.255.255.0
        - default-router 192.168.10.1
        - dns-server 192.168.10.100
        - exit
        - end
    2. Configure router interface as follows using CLI
        - enable
        - configure terminal
        - interface gigabitEthernet 0/0/0
        - ip address 192.168.10.1 255.255.255.0
        - no shutdown
        - exit
        - end
    3. Check the interface config using following command
        - show ip interface brief

4. Set PCs to DHCP

Activity 3:

Configuring DNS

- Objective: Introducing the concept of DNS and making students aware of how to manually configure DNS for a server
- Instructions:
    1. Configure the server with IP address of 192.168.10.100
    2. Then go to Services -> DNS and toggle it ON
    3. Create www.university.local and assign 192.168.10.100
    4. Check the DNS resolve from any PC using following command
        - Ping www.university.local

Activity 4:

Configuring HTTP

- Objective: Introducing the concept HTTP and making aware of how to configure HTTP manually on a server
- Instructions:
    1. On the server go to Services -> HTTP and toggle it ON
    2. Open web browser from any PC
    3. Try to visit http://www.university.local
    4. Use Simulation Mode to examine the communication

Activity 5:

Configuring FTP

- Objective: Introducing the concept FTP and making aware of how to enable FTP on a server manually
- Instructions:
    1. Enable FTP on server
    2. Create a user account
    3. Try to connect FTP from a PC using FTPclient